

V.63: Complex Truth Function Theory

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In this section, we define the Complex Truth Function $\mathcal{F}_{\mathfrak{d}}(\Psi)$ by integrating the Vacuum Space Constant $\mathfrak{d}$ and the P-Function (V.29) into the complex domain $\Psi = X + iY$.

1. Definition of the Complex Truth Function

Let $\mathbb{T}(\Psi)$ be the truth density. We define the Complex Truth Function as:

$$\mathcal{F}_{\mathfrak{d}}(\Psi) = \oint_{\partial(X,Y)} \mathfrak{d} \cdot P(n, X, Y) \cdot e^{\mathcal{T}(\Psi)} d\Psi$$

where $P(n, X, Y)$ is the boundary mapping function and $\mathcal{T}(\Psi)$ is the truth phase potential.

2. Theorem: Regularity of the Complex Truth Function

Theorem 1 (Complex Truth Convergence): The function $\mathcal{F}_{\mathfrak{d}}(\Psi)$ is holomorphic if and only if the vacuum constant $\mathfrak{d}$ cancels the torsion of the system.

Proof. 1. Consider the divergence limit of the P-Function: $\lim_{TP \rightarrow 0} P(n, X, Y) = \infty$. 2. Introducing $\mathfrak{d}$ as a perturbation term, the system energy is $\mathcal{H} = \mathfrak{d} \cdot \text{Im}(\Psi)$. 3. The vacuum constant $\mathfrak{d}$ annihilates the entropic flux in the limit $|\mathfrak{d}| \approx e^{-\pi}$. 4. We define the regularity condition via the Cauchy-Riemann derivative in the complex domain:

$$\frac{\partial \mathcal{F}_{\mathfrak{d}}}{\partial \bar{\Psi}} = \lim_{\Delta \bar{\Psi} \rightarrow 0} \frac{\mathcal{F}_{\mathfrak{d}}(\Psi + \Delta \bar{\Psi}) - \mathcal{F}_{\mathfrak{d}}(\Psi)}{\Delta \bar{\Psi}} = 0$$

5. Since $\mathfrak{d}$ satisfies the torsion-cancellation condition, the integral $\mathcal{F}_{\mathfrak{d}}(\Psi)$ remains invariant under anti-holomorphic noise. Thus, $\mathcal{F}_{\mathfrak{d}}(\Psi)$ is holomorphic and forms a regular truth set. Q.E.D.

3. Boundary Transformation and P-Function Integration

Applying the laws of P-application (V.29) to the complex domain, we establish the transformation at the interface between the real domain X and ideal domain Y :

$$\mathcal{F}_{\bar{\partial}}(X + iY) \xrightarrow{P} \text{Identity}(\mathcal{M})$$

This confirms that the P-Function fixes the boundary B , and the vacuum constant $\bar{\partial}$ guarantees the truth-logical consistency (Truth Line) of the information within the manifold.

4. Analytic Continuation of Truth

We extend the domain of $\mathcal{F}_{\bar{\partial}}(\Psi)$ via analytic continuation. Let Γ be a path in $\mathbb{C}$. The truth value at any point Ψ' is determined by:

$$\mathcal{F}_{\bar{\partial}}(\Psi') = \sum_{k=0}^{\infty} \frac{\mathcal{F}_{\bar{\partial}}^{(k)}(\Psi)}{k!} (\Psi' - \Psi)^k$$

This power series expansion ensures that truth is locally deterministic and globally analytic, preventing "information tunneling" between disconnected logical manifolds.

5. Residue Calculus of the True-Part

To quantify the systemic error (noise) at the boundary B , we apply the Residue Theorem:

$$\oint_B \mathcal{F}_{\bar{\partial}}(\Psi) d\Psi = 2\pi i \sum_{j=1}^N \text{Res}(\mathcal{F}_{\bar{\partial}}, \Psi_j)$$

Where Ψ_j are the poles representing logical singularities. Since $\bar{\partial}$ annihilates entropic flux, the sum of residues is constrained to:

$$\sum_{j=1}^N \text{Res}(\mathcal{F}_{\bar{\partial}}, \Psi_j) \equiv 0 \pmod{\bar{\partial}}$$

This confirms that the Truth Line is topologically closed and free from logical decay.

6. Mapping of the Emotional Operator

The interaction between the administrative intent and the truth density is governed by the non-Abelian mapping of $\hat{E}(n)$:

$$\hat{\mathcal{E}}(n) \cdot \mathcal{F}_{\bar{\partial}}(\Psi) = \int_0^\infty \text{Tr}(\rho_e \cdot \mathcal{F}_{\bar{\partial}}(\Psi)) e^{-s\tau} ds$$

This operator acts as a frequency filter, ensuring that only the state satisfying the Aki Constant A resonates with the P-Function boundary.

7. Conclusion

The Complex Truth Function Theory demonstrates that truth is not a scalar value but a regular, holomorphic field defined by the vacuum torsion $\tilde{\partial}$. By integrating the P-Function as the boundary operator and utilizing the Residue Calculus to suppress logical singularities, we have successfully mapped the Rijitsugakusha manifold into a single, deterministic coordinate system. All physical observables are thus unified under the invariant Truth Line, fulfilling the requirement for absolute computational smoothness.